

Recoverability-Gated Deliberation for Release-Aware Slow Reasoning in Closed-Loop Driving

Anonymous Authors

Abstract—Large language models can provide deliberative reasoning for autonomous driving, but inference latency separates the state that initiates a request from the state in which its response becomes available. Existing asynchronous designs preserve a fast control stream during inference, whereas deciding whether a delayed proposal should affect the current command remains a distinct runtime problem. This paper presents Recoverability-Gated Deliberation (RGD), a two-stage framework for slow-reasoner admission and response release. At query time, RGD selects slow requests using delay feasibility, action support, recovery-cost advantage, and scene demand. At release, it maps the returned proposal to the observed state and admits it only when it passes state checks, differs from the contemporaneous fast action, and satisfies the recovery-cost condition. We also introduce Recoverable Value of Deliberation (R-VoD), an offline matched-rollout diagnostic of the corrective opportunity retained at release. Experiments in highway-env examine controlled delays, selective invocation, response traces, and backend substitution. A zero-delay MetaDrive study examines transfer of the common action interface. The results show that RGD preserves fast-path continuity and allows a delayed proposal to affect the final projected command only after it satisfies the release contract.

Index Terms—Autonomous driving, closed-loop decision making, inference latency, large language models, recoverability, selective computation.

I. INTRODUCTION

LARGE language models (LLMs) and vision-language models (VLMs) have been used for semantic scene interpretation, driving-knowledge retrieval, and high-level decision making in autonomous driving systems [1]–[8]. These capabilities supplement reactive policies with semantic context and long-horizon intent. However, closed-loop deployment introduces a release-time mismatch: a reasoning request is issued from one traffic observation, while its proposal becomes available in a later state. A high-frequency controller can maintain continuous vehicle control while a deliberative model runs asynchronously, but the system must still decide whether the delayed proposal should influence the current command.

Consider a closing lead vehicle and an initially open adjacent lane. The system may request a lane-change recommendation from the slow reasoner. During inference, traffic evolves and the fast controller continues to act. When the response arrives, the target gap may have narrowed, the original maneuver may no longer be admissible, or the fast controller may already select the same or a preferable action. The release decision must therefore test the proposal against the observed state and the contemporaneous fast action.

Maintaining a fast control stream does not resolve this release decision. A returned action may pass an isolated state check yet coincide with the fast action or offer no recovery-cost advantage

after traffic evolves. Both branches therefore require a common action schema, mapper, and state-dependent evaluator. This motivates RGD’s separation of query admission from release validation.

Recent studies address relevant parts of this process. Invocation policies use uncertainty, scene conditions, or policy discrepancy to decide when to engage a slow branch [9]–[11]. VLM-based attention allocation prioritizes surrounding vehicles under behavioral uncertainty [12]. AsyncDriver and NetRoller preserve fast control during model inference [8], [13], while DualDrive aligns semantic intent with a real-time policy [14]. Runtime-assurance methods determine whether a candidate action may enter the actuation path [34]. These designs primarily address invocation timing, control continuity, or candidate safety. An explicit release-time comparison between a delayed proposal and the contemporaneous fast action through a shared action interface is generally not reported.

This paper addresses two questions in delayed closed-loop driving: whether the evolved traffic state still admits a corrective action and whether the returned proposal provides such an action at release. Two constraints guide the design: the fast controller must remain active throughout inference, and rejected proposals must not disrupt the control loop. Both proposals are evaluated at the same release state through a shared action mapper and feasibility interface.

RGD treats the slow reasoner as an optional proposal generator rather than a replacement for the fast controller. The fast action remains the default, and a slow proposal reaches actuation only through the shared mapper and safety interface. This separation allows the reasoner to be changed without modifying the release rule and makes the source of each executed action explicit.

A release state is *recoverable* when the remaining maneuver window contains at least one feasible effective alternative that remains distinct from the fast effective action and exceeds a matched-rollout utility margin. At query time, RGD applies online delay, support, cost, and demand conditions before issuing a slow request. *Actionability* is evaluated at release: the returned mapped command must pass the release-state checks, differ from the mapped fast command, and satisfy the recovery-cost margin. Recoverability therefore describes the release state, whereas actionability describes the returned proposal before final safety projection.

Recoverable Value of Deliberation (R-VoD) quantifies recoverability offline. It compares alternative and fast first actions through matched rollouts from the same logged release state under a common continuation policy. It therefore measures the corrective opportunity remaining at release rather than the

utility of a returned slow proposal. R-VoD is a reward-defined diagnostic, not an online value estimate, and does not enter admission or release validation.

The online gate requires scene demand, delay feasibility, action support, and cost advantage simultaneously; high urgency alone cannot compensate for an expired maneuver window or the absence of a supported alternative. The fast controller supplies every action during inference. When the response arrives, release validation evaluates the proposal in the current traffic state and accepts it only if it passes current-state mapping, state checks, distinctness from the fast command, and recovery-cost checks.

We evaluate RGD under a shared fast controller, slow reasoner, request budget, and action interface across controlled delay settings. Event-level response traces link each request to release validation and final projection. Component analysis assesses selected admission conditions, while backend substitution and adapter experiments examine the common action interface. Additional traffic sweeps characterize operation under the fixed interface.

The main contributions are as follows:

- 1) We formalize delayed deliberation through two release-time notions: recoverability of the current state and actionability of a returned proposal. We define R-VoD as an offline matched-rollout diagnostic of recoverability under a common continuation policy.
- 2) We present RGD, which combines noncompensatory query admission with a release contract aligned to the fast action. The fast controller remains active during inference, and a delayed proposal can affect control only after both branches are mapped and evaluated in the observed release state through the shared safety interface.
- 3) We conduct controlled-delay, trace-based, component, backend-substitution, and adapter-transfer evaluations. These studies examine selective invocation, release-contract enforcement, and use of the shared action interface under the evaluated settings.

II. RELATED WORK

A. Language-Conditioned and Generative Driving

Language-conditioned driving methods differ in the proximity of their outputs to vehicle actuation. At the advisory level, DiLu retrieves relevant experience before producing an interpretable decision [1], and Driving with LLMs represents traffic through object-level vectors and returns actions with explanations [15]. VLM-based systems combine visual observations with language reasoning for navigation and motion planning [6], [16], while knowledge retrieval methods supply domain-specific context before decision making [7]. These systems use language-conditioned interfaces to encode scene relations and retrieve driving knowledge.

A second group places model outputs closer to the control boundary. Policy transfer methods use language models to support vehicle decision making [17], GPT-Driver casts planning as language modeling [5], and C-TRAIL integrates commonsense reasoning with trajectory planning [18]. Generative driving

methods learn predictive scene representations or future trajectories for downstream planning [19]–[22]. As outputs approach actuation, a delayed proposal may have fewer downstream opportunities for correction and should be evaluated in the state in which it will execute.

These studies generally focus on model capability or driving representation. RGD instead addresses whether a delayed high-level proposal may replace the current fast action. It operates at the interface between a slow proposal generator and the fast controller, without introducing a new language model or trajectory decoder. The release rule evaluates the returned proposal in the state in which it may be executed.

B. Dual Process Reasoning and Selective Computation

Dual-process driving systems divide responsibility between a responsive policy and a deliberative model. LeapAD uses reflection and accumulated experience, LeapVAD adds cognitive perception, and FASIONAD coordinates both branches through adaptive feedback [9], [23], [24]. DualDrive aligns semantic intent with a real-time policy, and NetRoller connects general and specialized models through asynchronous execution [8], [14]. In each design, the fast branch supplies immediate control while the slow branch contributes information or actions at selected intervals.

Selective computation methods determine when the slow branch should run. AdaDrive and AdaThinkDrive adapt deliberation depth to scene demand [10], [25], and G-VLM calibrates invocation from the discrepancy between lightweight and multimodal policies [11]. Behavioral Attention uses a VLM to allocate attention among surrounding vehicles under behavioral uncertainty [12]. These invocation signals are evaluated before inference begins, when neither the returned action nor the release state is known. RGD supplements invocation selection with release-time evaluation against the contemporaneous fast action.

Query-time selection and release-time validation serve different purposes. The first determines whether slow computation should begin; the second determines whether its output remains useful after the state has changed. RGD retains both decisions explicitly rather than treating a successful request as sufficient for release.

Metareasoning balances expected decision improvement against computational cost. Time-bounded and anytime methods schedule reasoning under a finite budget [26]–[29], and concurrent planning permits an executor to continue acting while a planner computes [30]. In driving, an estimate made at query time can differ from the opportunity available at release because traffic evolves while the fast controller continues to act. R-VoD operationalizes the remaining corrective opportunity through matched rollouts.

C. Runtime Safety, Delay, and Viability

Timing mechanisms preserve control during inference. AsyncDriver separates language-model planning from real-time control [13], NetRoller keeps a specialized driving model active [8], and event-triggered or adaptive-computing strategies regulate update frequency [31], [32]. Delay-aware control

further models state evolution during input lag [33]. At the assurance layer, Simplex-based systems revert to a trusted controller when a learned action fails an acceptance test [34], and formal viability methods characterize admissible motion under dynamic constraints [35], [36].

These approaches address timing, invocation, or candidate safety. They do not generally formulate a comparison between a delayed proposal and the contemporaneous fast action as a separate release-time decision. RGD complements rather than replaces runtime safety mechanisms. The shared safety interface checks whether a mapped action can enter actuation, while release validation additionally tests whether the delayed proposal is distinct from, and has a recovery-cost advantage over, the contemporaneous fast action. A safe but equivalent proposal therefore leaves the fast action unchanged. The final safety shield remains active after selection.

III. RECOVERABILITY-GATED DELIBERATION

RGD separates a slow-reasoner call into query admission and release validation. Admission determines whether the scene warrants deliberation and whether a feasible alternative is supported under the expected delay. Release validation checks the returned proposal in the observed release state. The fast controller supplies the executed action while inference is pending and whenever either stage rejects the slow path.

A release state s_r is *recoverable* if it contains a feasible effective alternative that remains distinct from the fast effective action and exceeds its matched-rollout utility by at least ϵ . R-VoD measures this state-level opportunity offline. A returned proposal is *actionable at release* only if its mapped command passes the contemporaneous state checks, differs from the mapped fast command, and satisfies the proposal-specific recovery-cost margin. The final safety projection may still map an authorized command to the fast executed action. Thus, R-VoD and release validation serve complementary rather than interchangeable roles.

As shown in Fig. 1, the online path contains the two gates, whereas the offline branch constructs R-VoD labels from matched first-action rollouts. The fast controller remains active at every cycle. Driving memory can provide context after admission, but it does not alter either gate and is not used in the reported frozen runs.

The architecture imposes a non-interference contract on slow reasoning. A slow request never suspends the fast controller, and a missing response, mapping failure, or rejected release leaves the fast command selected. The reasoner is therefore a proposal source rather than a second controller. This boundary prevents inference latency and response-format errors from changing the control cadence.

The release comparison is action-aligned. Both fast and slow proposals first enter the same mapped command space, after which the same feasibility and cost interfaces are applied. Semantically equivalent language outputs are therefore treated as the same maneuver, and backend-specific wording cannot create an artificial action difference. The shared final safety shield remains the last stage for either source.

A. Query and Release Process

At frame t , the ego vehicle observes state s_t . The fast policy π_f is mapped by the shared action mapper to a_t^f . Both branches use this mapper and the same final safety projection $\Phi(s, a)$, which projects a mapped command to its final executable action in state s . The mapper converts each proposal into a command and target speed; different expressions are equivalent when they induce the same mapped behavior. The high-level action domain comprises lane-left, lane-hold, lane-right, accelerate, and decelerate, all mapped through the same simulator interface.

The returned proposal is mapped in the observed release state s_r . A lane-change or speed command available at query time may no longer be valid; a mapping or feasibility failure leaves a_r^f selected.

The delay estimate is used only for admission. The actual release frame r is the first control frame at which the response is available, and returned and fast actions are evaluated in the observed state s_r . Responses unavailable before episode termination are not released.

B. Offline Measure of Corrective Opportunity

Offline matched rollouts quantify the corrective opportunity at release. At each logged release snapshot s , let $a_{\text{eff}}^f = \Phi(s, a^f)$ denote the effective fast first action after the shared mapper and final safety projection. $\mathcal{D}(s)$ contains feasible effective first actions that remain distinct from a_{eff}^f after the same stages, and $\mathcal{C}(s) = \{a_{\text{eff}}^f\} \cup \mathcal{D}(s)$. The matched-rollout utility and Recoverable Value of Deliberation (R-VoD) are

$$J(s, a) = \frac{\sum_{k=0}^{K-1} \gamma^k \bar{r}_k(s, a)}{\sum_{k=0}^{K-1} \gamma^k} - \lambda_{\text{col}} \mathbb{I}_{\text{col}}(s, a), \quad (1)$$

$$\text{R-VoD}(s) = \max_{a \in \mathcal{C}(s)} [J(s, a) - J(s, a_{\text{eff}}^f)].$$

Here, K is the common positive rollout horizon, $\gamma \in (0, 1]$ is the discount factor, $\bar{r}_k$ is the normalized simulator reward, and $\lambda_{\text{col}} > 0$ weights the collision indicator $\mathbb{I}_{\text{col}}(s, a)$. Reward increments after termination are set to zero. Only the first effective action is changed; the fast policy supplies the continuation. Because $a_{\text{eff}}^f \in \mathcal{C}(s)$, R-VoD is nonnegative and equals zero when no alternative improves the matched fast continuation. A state is labeled corrective when $\text{R-VoD}(s) \geq \epsilon$, where $\epsilon > 0$ is a fixed diagnostic margin.

The matched design fixes the logged release state, traffic realization, maximum horizon, continuation policy, and termination rule. Accordingly, $J(s, a)$ is an offline, reward-defined utility for the diagnostic; it is not queried by the online gate. By contrast, $c_t(a)$ is a runtime recovery-cost score used to rank current actions at admission and to test a returned command at release. The two quantities are purpose-built for different decisions and are not interchangeable.

Changing only the first effective action isolates the corrective opportunity available in the release state under the common simulator continuation. R-VoD therefore does not reward response length, explanation quality, or a particular LLM backend. It supplies a state-level label against which an admission policy can be evaluated. Release validation tests whether a returned proposal satisfies the online release contract; the final projection separately determines its executed effect.

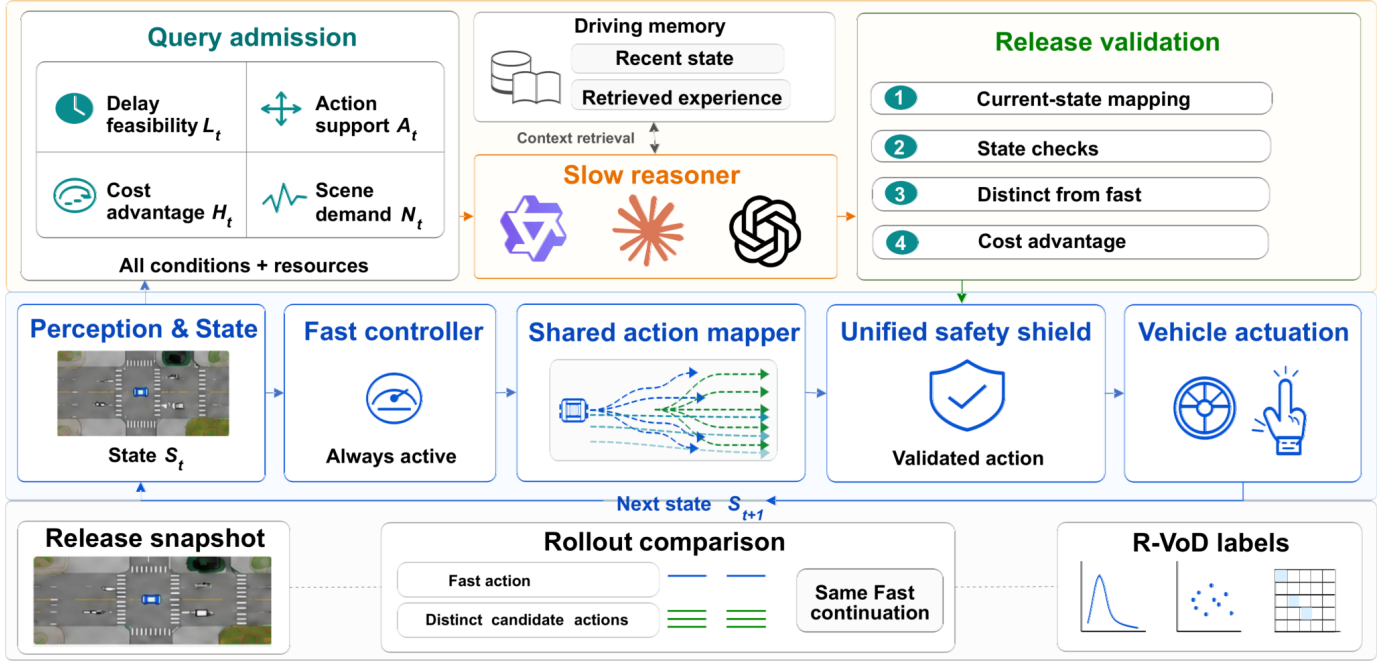


Fig. 1. RGD architecture and information flow. Query admission requires delay feasibility (L_t), action support (A_t), cost advantage (H_t), scene demand (N_t), and resource availability. At release, a returned proposal is evaluated through current-state mapping, recomputed state checks, mapped-command distinctness from the fast branch, and a proposal-specific cost margin before the shared safety projection. The optional driving-memory block was disabled in the frozen evaluations. Matched first-action rollouts from release snapshots provide offline R-VoD labels.

C. Query Admission

The online gate evaluates four conjunctive conditions. All must hold; high scene demand cannot compensate for an expired maneuver window.

Let $\mathcal{A}_t$ denote the current mapped action set. The fixed evaluator assigns each member a normalized recovery cost $c_t : \mathcal{A}_t \rightarrow [0, 1]$, where lower values are preferred, and a normalized support $u_t(a) \in [0, 1]$. Let $\mathcal{D}_t^{\text{alt}}$ contain feasible actions whose mapped commands differ from a_t^f . For each alternative action family m , its support $\bar{u}_{t,m}$ is the largest $u_t(a)$ among feasible members; an unavailable family contributes zero. Action support A_t is the mean of these family-wise maxima and is set to zero when no alternative family is available or the evidence is incomplete. Let $c_t^{\text{alt}} = \min_{a \in \mathcal{D}_t^{\text{alt}}} c_t(a)$ when the set is nonempty and $c_t^{\text{alt}} = c_t(a_t^f)$ otherwise.

The remaining gate factors are

$$\begin{aligned} L_t &= \text{clip}\left(1 - \frac{\hat{\ell}_t + \rho_t}{\max(d_t, \varepsilon_d)}, 0, 1\right), \\ H_t &= \text{clip}\left(\frac{[c_t(a_t^f) - c_t^{\text{alt}}]_+}{\kappa}, 0, 1\right), \\ N_t &= \max\{n_t^{\text{haz}}, n_t^{\text{int}}\}. \end{aligned} \quad (2)$$

Here, L_t is delay feasibility, H_t is cost advantage, and N_t is scene demand. The terms $n_t^{\text{haz}}, n_t^{\text{int}} \in [0, 1]$ are normalized hazard and interaction evidence, respectively. The timing quantities $d_t, \hat{\ell}_t, \rho_t \geq 0$, and $\varepsilon_d > 0$ use the same time unit; d_t is the estimated time to the earliest active maneuver constraint, ρ_t is a timing reserve, and ε_d is a timing regularizer. The positive constant κ normalizes the recovery-cost advantage.

A bounded planning horizon replaces d_t when no finite constraint is active. The operator $[x]_+ = \max\{x, 0\}$, and $\text{clip}(x, 0, 1) = \min\{1, \max\{0, x\}\}$. If $\mathcal{D}_t^{\text{alt}}$ is empty, $H_t = 0$. The maximum in N_t avoids counting correlated hazard and interaction evidence twice.

The factors encode different necessary conditions. L_t tests whether computation fits within the maneuver window, A_t requires support across alternative action families, H_t requires at least one lower-cost alternative, and N_t establishes a reason to deliberate. Their conjunction is deliberately noncompensatory: high scene demand cannot offset expired timing, and broad action support cannot offset the absence of a cost advantage. This structure preserves the operational meaning of each gate factor across delay settings.

Resource availability, including the request budget, cooldown, and executor status, is represented by $v_t \in \{0, 1\}$; $v_t = 1$ also requires that no slow request is pending. The admission rule is

$$q_t = \mathbb{I}[v_t = 1 \wedge L_t \geq \tau_L \wedge A_t \geq \tau_A \wedge H_t \geq \tau_H \wedge N_t \geq \tau_N], \quad (3)$$

where $\mathbb{I}[\cdot]$ is the indicator function and $\tau_j \in (0, 1]$, $j \in \{L, A, H, N\}$, are generic gate floors selected on development scenarios and fixed before evaluation.

If latency evidence, the action-cost vector, or the slow executor is unavailable, the corresponding condition is false and the fast controller continues without issuing a slow request.

D. Release Validation

At the actual release frame r , let y_r be the returned response and $\mathcal{M}$ the shared current-state mapper. The fast policy produces the mapped fast command a_r^f from the same state.

If $\mathcal{M}(s_r, y_r)$ is undefined, the release gate is set to zero and the fast command is retained. Otherwise, let $\tilde{a}_r^{\text{sl}} = \mathcal{M}(s_r, y_r)$ denote the mapped slow command. Only A_r , H_r , and N_r are recomputed from s_r ; L_t is used only for query admission. Let $\mathcal{V}_r \subseteq \mathcal{A}_r$ denote the feasible mapped commands when these release-state conditions hold, and let $\mathcal{V}_r = \emptyset$ otherwise. If a condition fails or $\tilde{a}_r^{\text{sl}} \notin \mathcal{V}_r$, the fast command is retained. Conditional on these checks, the remaining test in Fig. 1 is

$$g_r = \mathbb{I}[\tilde{a}_r^{\text{sl}} \neq a_r^f \wedge c_r(a_r^f) - c_r(\tilde{a}_r^{\text{sl}}) \geq \tau_R]. \quad (4)$$

Here, $\tau_R \in (0, 1]$ is a generic release margin under $c_r(\cdot) \in [0, 1]$ and is fixed before evaluation.

The mapped slow command is selected when $g_r = 1$; otherwise, the fast command is retained. The shared safety projection Φ then produces the executed action. Distinctness is tested between mapped commands before Φ , so the shield may still map an authorized slow command to the same executed command as the projected fast action. Post-projection distinctness is therefore an execution outcome reported separately. The release-state H condition establishes that a lower-cost alternative exists, while the final term in (4) tests whether the returned mapped command meets the required margin. No R-VoD label enters this online decision.

Release validation is stricter than a slow-action safety test. Feasibility asks whether the mapped proposal can execute in s_r , distinctness asks whether it differs from the contemporaneous mapped fast command, and the final comparison asks whether it satisfies the recovery-cost margin. A proposal admitted at query time receives no persistent authority; all response-specific checks use the observed release state.

IV. EXPERIMENTS AND DISCUSSION

A. Experimental Setup

We evaluate RGD on the highway-env [37] and MetaDrive [38] closed-loop simulators with Qwen3-8B [39] as the default slow reasoner. The main highway-env experiments operate at 10 Hz for 30 s. Each comparison shares a fixed seed cohort, action schema, fast controller, and evaluation rule; gate development, main evaluation, and mechanism analysis use separate cohorts. The main analyses use 30 seeds; the component analysis uses a separate 20-seed cohort. Unless stated otherwise, intervals denote 95% percentile bootstrap confidence intervals with simulator seed as the resampling unit. R-VoD is computed only after release through matched first-action rollouts. Gate floors, normalization constants, and corrective margins are selected on the development cohort and remain fixed throughout the reported evaluation.

The experiment suite gives each analysis a distinct evidentiary role. System-level context places the closed-loop implementation in established task settings. The delay sweep tests whether admission allocates slow computation according to the available maneuver window, while the response trace tests the release contract after the traffic state evolves. The component study examines how selected admission conditions shape the states sent to the slow path. Backend substitution evaluates the common action interface under different reasoners,

and the traffic and simulator studies characterize the fixed interface across evaluated road structures and adapters. This organization links each result to a specific system property.

B. System-Level Context

Following the PADriver experimental conditions [40], we evaluate RGD and report the resulting values alongside source-reported context. The source entries retain their original definitions and evaluation scope; Table I is therefore not a statistical head-to-head comparison. Within this protocol context, RGD completes 29 of 30 runs, records 611 m and 73.60 km/h, and operates at 37.6 ms per frame without a dedicated training phase on the evaluation environment. The fast controller remains active at every frame, whereas the slow reasoner is invoked selectively. Saf. and Kep. denote safe-distance keeping and lane keeping, respectively.

Hu et al. [6] define success as 30 consecutive collision-free frames. We adopt their three density settings and ten seeds per setting with no added delay. Fig. 2 reports success rates alongside published values, and Table II provides RGD's operating metrics under the settings stated in the source paper.

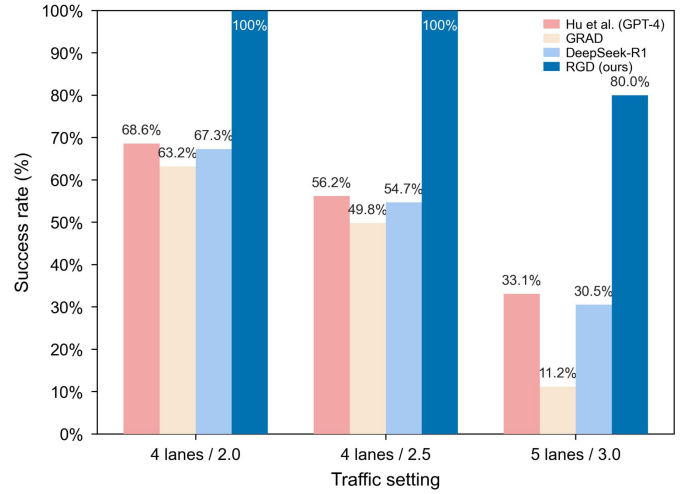


Fig. 2. Success-rate context in the highway-env settings of Hu et al. GPT-4, GRAD, and DeepSeek-R1 are source-reported values; their placement with RGD is not a head-to-head statistical comparison.

Under the settings stated by Hu et al., RGD attains 100% success at 4L/2.0 and 4L/2.5. At 5L/3.0, it attains 80% success and schedules 0.8 slow-path requests per episode. Table II uses safety-qualified speed, assigns zero to unsuccessful episodes, and averages requests over the ten episodes in each setting. The per-frame runtime remains below 66 ms in all three settings.

C. Delay-Controlled Allocation and Component Evidence

1) *Zero Added Delay Reference*: The zero-delay experiment provides a reference in which all six allocation policies share the same fast controller, slow reasoner, request budget, and feasibility interface; only the request and release rules differ. The subsequent delay sweep uses the same seeds to examine how delay feasibility changes slow-path allocation.

Without added delay, Fig. 3(a)–(b) shows that RGD schedules fewer slow-path requests and has a lower per-frame runtime

TABLE I
PROTOCOL-CONTEXT VALUES IN THE PADRIVER SETTING (NOT A HEAD-TO-HEAD COMPARISON).

Method	Distance (m)	Speed (km/h)	Saf.	Kep.	Completed	Runtime (s/frame)
Driving with LLMs	411	49.42	0.49	0.56	20	48.0
GPT-Driver	428	51.40	0.45	0.52	23	36.2
DiLu	386	46.29	0.60	0.72	28	24.1
PADriver-Normal	603	72.47	0.91	0.92	25	1.4
RGD (ours)	611	73.60	0.75	0.86	29	0.0376

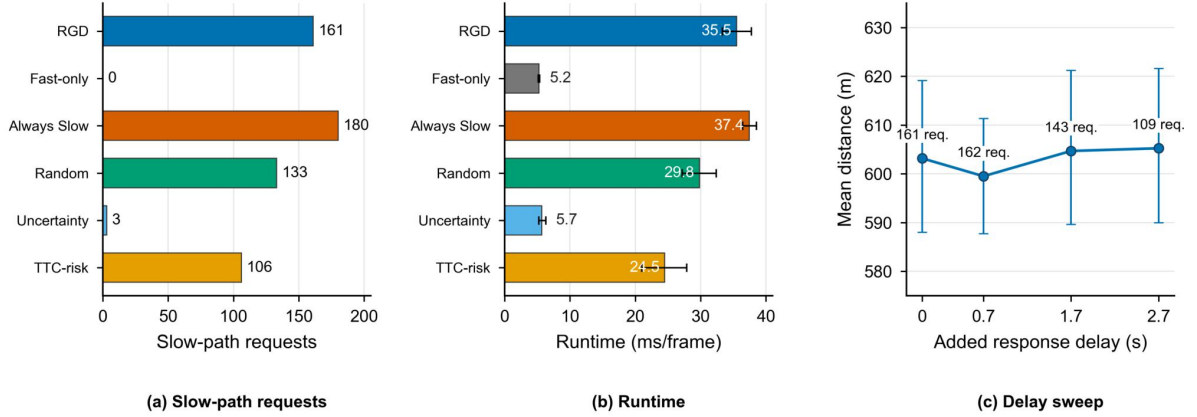


Fig. 3. Zero-delay policy comparison and RGD delay sweep: (a) slow-path request totals, (b) per-frame runtime, and (c) distance and requests versus added delay. Request counts aggregate 30 seeds; error bars denote 95% seed-bootstrap confidence intervals for runtime and distance.

TABLE II
RGD OPERATING METRICS UNDER THE SOURCE PROTOCOL CONTEXT OF HU ET AL.

Setting	Dist. (m)	Speed (m/s)	Requests/ep.	Runtime (ms/frame)
4L / 2.0	69.29	23.84	1.0	58.9
4L / 2.5	66.19	22.76	1.0	65.2
5L / 3.0	52.38	17.38	0.8	50.1

than Always Slow. This result shows that query admission reduces continual slow-path use under the same fast controller and slow reasoner.

Across the delay sweep in Fig. 3(c), distance remains within overlapping confidence intervals while the request count decreases at longer added delays. This allocation pattern is consistent with the delay-feasibility condition becoming more selective as the response interval grows. The fixed gate therefore adjusts its invocation rate without per-delay retuning.

2) *Allocation With a 1.7-s Added Delay*: The 1.7-s experiment examines how RGD processes a slow response after the traffic state has evolved. The fast controller, slow reasoner, request budget, seed cohort, and feasibility interface remain fixed. Fig. 4 reports the response-validation stages and the resulting release decisions.

Table III reports a separate 30-seed backend evaluation under the 1.7-s setting. “Calls” denotes successful service responses rather than request attempts in the main cohort.

The fixed delay ensures that each returned proposal is evaluated in a state later than the state that initiated the request. RGD therefore applies a returned proposal only after all release checks.

TABLE III
BACKEND SUBSTITUTION UNDER A FIXED RGD INTERFACE AT A 1.7-S ADDED DELAY.

Reasoner	Calls	Distance (m)	Completion
Qwen3-8B	138	599.16	0.97
Qwen3.5-4B	138	599.16	0.97
Qwen2.5-7B	122	599.12	0.97
GPT-5.6-sol	112	603.39	0.97
GPT-5.6-terra	132	599.16	0.97
GPT-5.6-luna	133	599.16	0.97
Fable-5	137	599.16	0.97

3) *Response Validation and Component Effects*: Response traces follow each request from admission through response validation, current-state mapping, and actuation. At the same 1.7-s added delay, the leave-one-out variants use a separate 20-seed cohort; each removes one admission condition while retaining the release procedure.

Under the 1.7-s delay, 143 RGD requests return valid responses and 138 reach release before episode termination. Current-state checks admit 28 responses to the release comparison; nine mapped slow commands differ from the contemporaneous fast command, and six satisfy the recovery-cost margin. After the shared final projection Φ , four of these authorized commands match the projected fast command and two remain distinct. Fig. 4(a) reports the complete event funnel, while panel (b) summarizes all 28 state-valid outcomes. Thus, a delayed proposal can affect the final command only after it satisfies the release contract and remains distinct after projection; otherwise, the fast action remains in control.

Fig. 5(a)–(b) shows that removing L , A , or H changes both

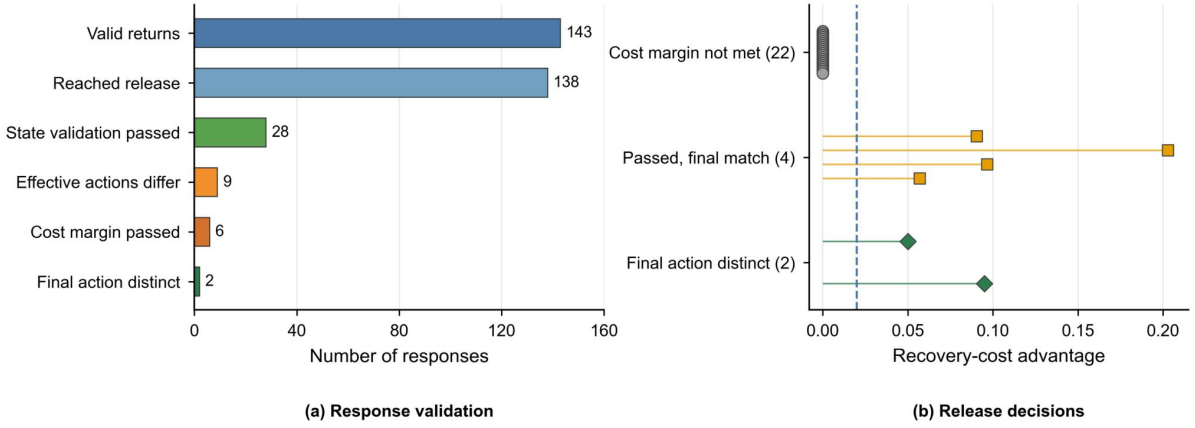


Fig. 4. Response-validation funnel at a 1.7-s added delay: (a) returned, release-reached, state-valid, mapped-distinct, cost-qualified, and projection-distinct responses; (b) recovery-cost advantages and final outcomes for the 28 state-valid responses. Stage counts aggregate the 30-seed main cohort; the dashed line denotes the release margin τ_R .

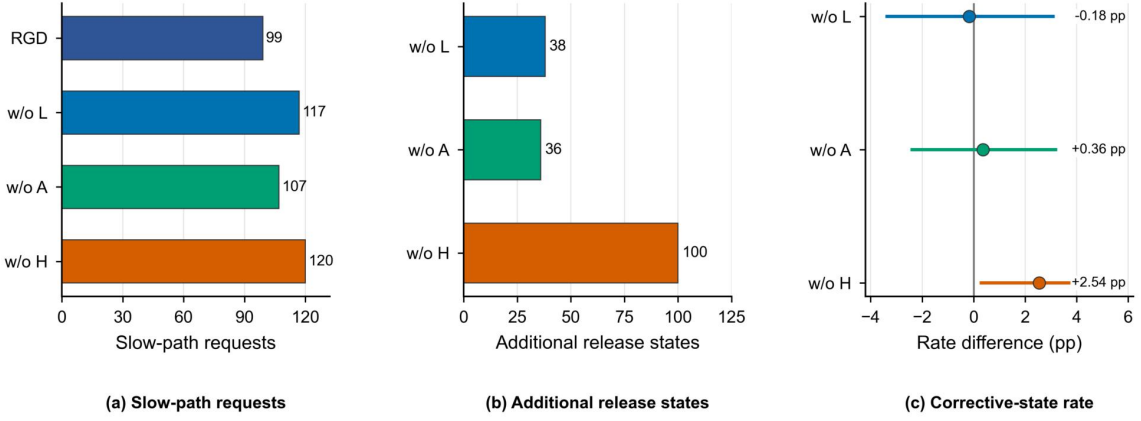


Fig. 5. Admission-component analysis: (a) slow-path request totals, (b) additional release states selected by the L , A , or H ablation, and (c) R-VoD-labeled corrective-state-rate differences computed as Full RGD minus the corresponding ablation. Counts aggregate the separate 20-seed mechanism cohort; error bars denote 95% confidence intervals.

the number of requests and the release states selected by the gate. Panel (c) reports the R-VoD-labeled corrective-state-rate difference between Full RGD and each ablation with 95% seed-bootstrap confidence intervals. The H contrast has a confidence interval entirely above zero, directly linking the online cost-advantage condition to the corrective-state composition of the selected cohort. The selection changes in panels (a)–(b) further show that L and A shape which states are sent to the slow path. Together, these analyses establish the complementary roles of delay feasibility, action support, and recovery-cost advantage in the admission rule.

The corrective-state rate in Fig. 5(c) is labeled offline by R-VoD, whereas H_t is computed online from the recovery-cost evaluator c_t . The two quantities serve different roles: H_t ranks current candidate actions before a request is issued, whereas R-VoD labels the opportunity that remains at the observed release state through matched rollouts. The H contrast therefore connects an online admission signal with a separately constructed release-state diagnostic.

D. Reasoner-Interface Evaluation

We replace only the slow reasoner while retaining the RGD gate, action schema, and fallback path. Table III reports closely aligned distance and completion values across the seven evaluated backends, from local models to API services. Successful service responses range from 112 to 138, whereas mean distance ranges from 599 to 603 m and every backend attains a completion rate of 0.97. All backends use the same action mapping, current-state validation, and recovery-cost comparison. Under this shared interface, the observed system-level point estimates remain closely aligned across the evaluated backends.

E. Evaluation Across Traffic and Simulators

1) *Traffic Structure*: The auxiliary sweep includes the default 4L/2.0 setting and five lane/density variants without retuning the gate. Table IV reports mean distance and successful runs over 30 episodes per setting. At density 2.0, successful runs range from 28 to 30 and mean distance from 599 to 629 m. Across all six cells, the fixed configuration completes 20–30

TABLE IV
RGD OUTCOMES UNDER LANE AND DENSITY SETTINGS.

Lanes/Density	Distance (m)	Successful runs (/30)
4L/2.0	599	29
4L/3.0	478	23
5L/2.0	629	30
5L/3.0	429	20
6L/2.0	611	28
6L/3.0	453	20

TABLE V
COMPOUND-HAZARD TASK CONTEXT IN METADRIVE
(SOURCE-CONTEXT COMPARISON).

Method	Scene	Safety (%)	Speed (m/s)
ASR-RL [17]	A	99.2	7.1
	B	98.3	8.6
	C	97.9	6.8
RGD (ours)	A	100.0	4.8
	B	94.8	7.0
	C	84.6	6.9

TABLE VI
MANEUVER AND TRAFFIC-DENSITY TASK CONTEXT (SOURCE-CONTEXT COMPARISON).

Method	Traffic Scenario					Left Turn Traffic Density		
	Left	Right	Straight	Merge	Round.	1.0	2.0	3.0
VLM attention allocation [12]	0.966	0.991	0.984	0.987	0.974	0.975	0.961	0.948
RGD (ours)	0.842	0.908	0.868	0.986	0.747	0.906	0.763	0.601

runs. These results describe operation of the same interface under the evaluated traffic structures.

2) *Simulator Transfer*: The MetaDrive experiment introduces simulator-specific observation and action adapters at zero response delay, with no modification to the gate configuration used in highway-env. The shared action schema, admission rule, and fallback order are applied without retuning. This study evaluates transfer of the common action interface across heterogeneous simulator dynamics; delayed release-time behavior is characterized by the highway-env experiments.

Fig. 6 visualizes representative frozen cross-simulator traces. The upper row contains highway-env highway, merge, roundabout, and intersection states; the lower row contains the corresponding MetaDrive state reconstructions. The paired layouts expose the same discrete high-level action schema in straight-road, merging, circulating, and cross-traffic geometries. The scenes provide visual context for the quantitative transfer studies.

Road geometry, traffic representation, and simulator dynamics differ across the two rows, yet each panel uses the same mapper, admission rule, and fast-path fallback. The zero-delay MetaDrive study exercises the common release interface through the simulator adapter. The MetaDrive panels reconstruct the logged ego state and recorded task-relevant neighbors at the selected frame, providing a direct view of the adapter boundary while preserving the traffic configuration used by the frozen evaluation.

F. Evaluation on Additional Scenario Families

The published ASR-RL and VLM-attention rows retain their source definitions and evaluation scopes, whereas the RGD rows are measured in this work. The values provide task-level context rather than a common-protocol statistical ranking.

1) *Compound Hazard Scenes*: Following the experimental conditions stated by ASR-RL [17], the MetaDrive evaluation covers pedestrian crossing (A), uncontrolled intersection (B), and combined vehicle-pedestrian traffic (C), each with 1,000 episodes. Table V reports the RGD results alongside source-reported ASR-RL context. RGD attains 100% collision-free

safety in Scene A and applies the same gate configuration across the three compound-hazard scenes without policy-library training or scenario-specific retuning.

2) *Maneuver and Density Changes*: Following the settings stated in the source paper, five maneuver classes and a left-turn density sweep are evaluated at 2 Hz for 30 s with 1,000 episodes per cell. Table VI reports the RGD success rates alongside published VLM attention-allocation values [12] as task-level context. RGD achieves 0.986 on merge, and the same high-level action interface and gate logic are applied across highway, junction, and curved-road maneuver families.

G. Discussion

RGD separates three related decisions. Before computation, admission asks whether a supported lower-cost alternative can remain useful within the expected response interval. After state evolution, release validation asks whether the returned proposal satisfies the contemporaneous contract relative to the action the fast controller would now take. The final safety projection then determines the executed command. This organization distinguishes recoverability of the release state, authorization of the returned proposal, and post-projection execution outcomes.

The controlled-delay studies characterize the admission decision, and the response trace evaluates the release contract. The delay sweep shows that the fixed admission rule changes slow-path allocation as the response interval changes. The trace follows each proposal through current-state mapping, recomputed state checks, mapped-command distinctness, recovery-cost comparison, and the shared safety projection. It therefore makes the fast path's default role explicit whenever the release contract is not satisfied.

The component analysis connects the admission rule to release-state evidence. The positive H contrast links explicit comparison with the fast action to the R-VoD-labeled composition of the selected cohort. Backend substitution, the traffic sweep, and cross-simulator studies address a separate system property: the common action interface can be used with different reasoner outputs, road structures, and simulator adapters under the evaluated settings.

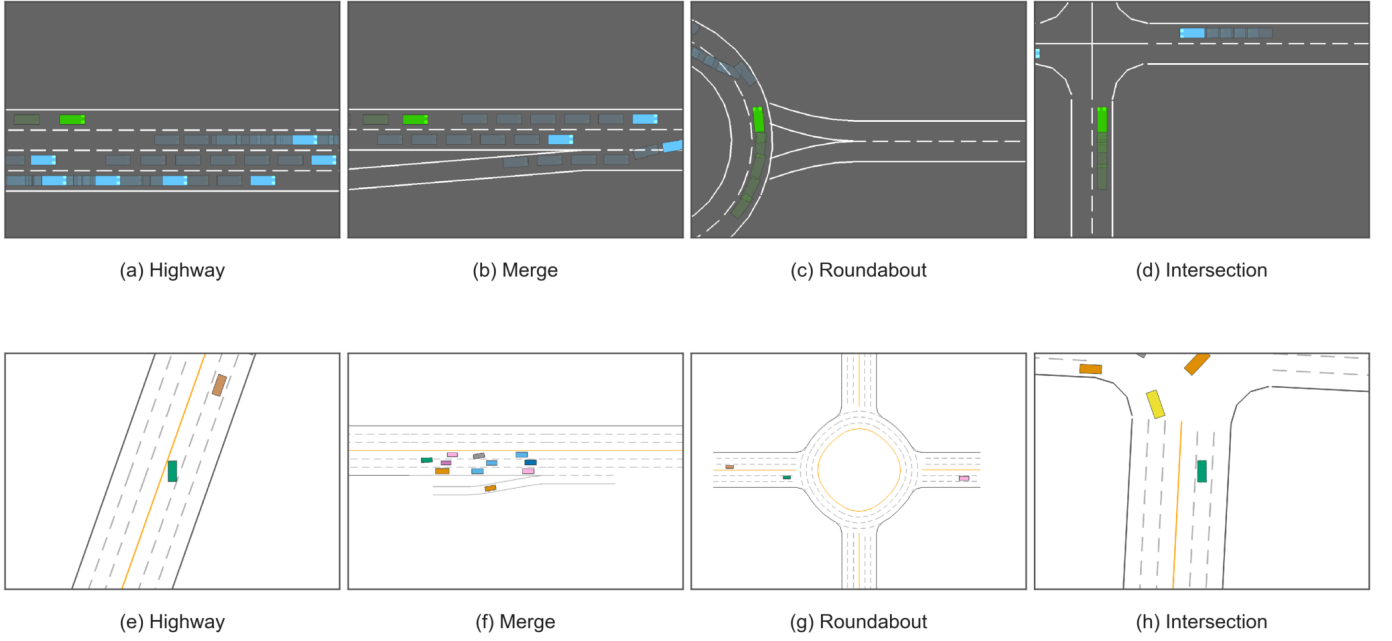


Fig. 6. Representative frozen closed-loop states under the shared RGD interface. The upper row shows highway-env: (a) highway, (b) merge, (c) roundabout, and (d) intersection. The lower row shows recorded-state reconstructions of the corresponding MetaDrive scenes: (e) highway, (f) merge, (g) roundabout, and (h) intersection. The ego vehicle is shown in green in every panel.

a) Scope and limitations.: The present study considers discrete high-level commands, controlled added delays, a fixed fast-policy continuation for R-VoD, and the evaluated simulator adapters. MetaDrive evaluates interface transfer at zero response delay. Extending the release contract to continuous trajectory proposals and measured service latency is left for future work.

V. CONCLUSION

This paper presents Recoverability-Gated Deliberation, a two-stage framework that determines when to invoke slow reasoning and whether a delayed LLM proposal may enter the control loop. By separating query admission from release validation, RGD invokes slow reasoning only when the joint delay, support, cost, and demand conditions hold. At release, it authorizes a returned mapped command only when it passes the state checks, differs from the contemporaneous fast command, and satisfies the recovery-cost margin. R-VoD provides an offline matched-rollout diagnostic of the corrective opportunity available at release.

Controlled-delay experiments, response tracing, and component analysis demonstrate the complementary roles of query admission and release validation. The response trace shows that a delayed proposal can affect the final command only after it satisfies the release contract and remains distinct after the shared final projection, while the fast action remains in control otherwise. Within the evaluated settings, the common RGD interface is exercised across LLM backends, traffic structures, and simulator adapters. By treating slow reasoning as an expiring proposal rather than a persistent command, RGD establishes a release-oriented decision layer between low-rate deliberation and high-frequency control.

REFERENCES

- [1] L. Wen *et al.*, “DiLu: A knowledge-driven approach to autonomous driving with large language models,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2024.
- [2] Z. Xu *et al.*, “DriveGPT4: Interpretable end-to-end autonomous driving via large language model,” *IEEE Robot. Autom. Lett.*, vol. 9, no. 10, pp. 8186–8193, 2024.
- [3] H. Shao *et al.*, “LMDrive: Closed-loop end-to-end driving with large language models,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 15 120–15 130.
- [4] C. Sima *et al.*, “DriveLM: Driving with graph visual question answering,” in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2024, pp. 256–274.
- [5] J. Mao, Y. Qian, J. Ye, H. Zhao, and Y. Wang, “GPT-Driver: Learning to drive with GPT,” arXiv preprint arXiv:2310.01415, 2023. [Online]. Available: <https://arxiv.org/abs/2310.01415>
- [6] Y. Hu, D. Ou, J. Huang, M. Wu, M. Hao, and R. Yu, “Integrating vision and language foundation models for enhanced navigation and decision-making in connected autonomous vehicles,” *IEEE Trans. Veh. Technol.*, vol. 74, no. 10, pp. 16 233–16 249, 2025.
- [7] D. Pei, J. He, K. Liu, Y. Wu, Y. Lei, and X. Xiao, “Enhancement of large language models driving knowledge for practical autonomous driving decision making,” *IEEE Trans. Veh. Technol.*, pp. 1–17, 2026, early access.
- [8] R. Xin *et al.*, “NetRoller: Interfacing general and specialized models for end-to-end autonomous driving,” *IEEE Trans. Veh. Technol.*, vol. 75, no. 5, pp. 7469–7482, 2026.
- [9] K. Qian *et al.*, “FASIONAD: Fast and slow fusion thinking systems for human-like autonomous driving with adaptive feedback,” arXiv preprint arXiv:2411.18013, 2024. [Online]. Available: <https://arxiv.org/abs/2411.18013>
- [10] R. Zhang *et al.*, “AdaDrive: Self-adaptive slow-fast system for language-grounded autonomous driving,” in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2025, pp. 5112–5121.
- [11] X. Chen, X. Wang, W. Chen, and J. Gao, “An adaptive multimodal end-to-end autonomous driving framework based on behavior discrepancy,” *IEEE Trans. Intell. Transp. Syst.*, pp. 1–12, 2026, early access.
- [12] B. Ma, H. Liu, R. Zhong, P. Liu, X. Zhou, and J. Ma, “Behavioral uncertainty-aware attention allocation via VLMs for interactive autonomous driving,” *IEEE Trans. Veh. Technol.*, pp. 1–14, 2026, early access.
- [13] Y. Chen, Z.-H. Ding, Z. Wang, Y. Wang, L. Zhang, and S. Liu, “Asynchronous large language model enhanced planner for autonomous driving,” in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2024, pp. 22–38.

- [14] L. Li, J. Fang, J. Xue, and C. Lv, "Vision-language model-enabled dual-system for autonomous driving in safety-critical transportation," *IEEE Trans. Intell. Transp. Syst.*, pp. 1–12, 2026, early access.
- [15] L. Chen *et al.*, "Driving with LLMs: Fusing object-level vector modality for explainable autonomous driving," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2024, pp. 14 093–14 100.
- [16] R. Zhao *et al.*, "VLM-Driver: Human-like autonomous driving decision-making via vision language model," *IEEE Trans. Veh. Technol.*, vol. 75, no. 5, pp. 7327–7342, 2026.
- [17] J. Wang, H. Ren, X. Zhu, and Z. Ma, "Enhancing autonomous vehicle decision-making through policy transfer with large language model," *IEEE Trans. Intell. Transp. Syst.*, pp. 1–10, 2025, early access.
- [18] Z. Cui *et al.*, "C-TRAIL: A commonsense world framework for trajectory planning in autonomous driving," *IEEE Trans. Veh. Technol.*, pp. 1–14, 2026, early access.
- [19] X. Wang, Z. Zhu, G. Huang, X. Chen, J. Zhu, and J. Lu, "DriveDreamer: Towards real-world-drive world models for autonomous driving," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2024, pp. 55–72.
- [20] C. Min *et al.*, "DriveWorld: 4D pre-trained scene understanding via world models for autonomous driving," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 15 522–15 533.
- [21] W. Zheng, R. Song, X. Guo, C. Zhang, and L. Chen, "GenAD: Generative end-to-end autonomous driving," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2024, pp. 87–104.
- [22] B. Liao *et al.*, "DiffusionDrive: Truncated diffusion model for end-to-end autonomous driving," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2025, pp. 12 037–12 047.
- [23] J. Mei *et al.*, "Continuously learning, adapting, and improving: A dual-process approach to autonomous driving," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 37, 2024, pp. 123 261–123 290.
- [24] Y. Ma *et al.*, "LeapVAD: A leap in autonomous driving via cognitive perception and dual-process thinking," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 37, no. 4, pp. 1963–1977, 2026.
- [25] Y. Luo *et al.*, "AdaThinkDrive: Adaptive thinking via reinforcement learning for autonomous driving," arXiv preprint arXiv:2509.13769, 2025. [Online]. Available: <https://arxiv.org/abs/2509.13769>
- [26] S. Russell and E. Wefald, "Principles of metareasoning," *Artif. Intell.*, vol. 49, no. 1–3, pp. 361–395, 1991.
- [27] M. Boddy and T. L. Dean, "Deliberation scheduling for problem solving in time-constrained environments," *Artif. Intell.*, vol. 67, no. 2, pp. 245–285, 1994.
- [28] S. Zilberstein, "Using anytime algorithms in intelligent systems," *AI Mag.*, vol. 17, no. 3, pp. 73–83, 1996.
- [29] B. Cserna, W. Ruml, and J. Frank, "Planning time to think: Metareasoning for on-line planning with durative actions," in *Proc. Int. Conf. Autom. Plan. Scheduling (ICAPS)*, vol. 27, 2017, pp. 56–60.
- [30] A. Elboher, A. Bensoussan, E. Karpas, W. Ruml, S. S. Shperberg, and E. Shimony, "A formal metareasoning model of concurrent planning and execution," in *Proc. AAAI Conf. Artif. Intell.*, vol. 37, no. 10, 2023, pp. 12 427–12 435.
- [31] P. Tabuada, "Event-triggered real-time scheduling of stabilizing control tasks," *IEEE Trans. Autom. Control*, vol. 52, no. 9, pp. 1680–1685, 2007.
- [32] M. Li, J. Gao, L. Zhao, and X. Shen, "Adaptive computing scheduling for edge-assisted autonomous driving," *IEEE Trans. Veh. Technol.*, vol. 70, no. 6, pp. 5318–5331, 2021.
- [33] T. G. Molnar, A. K. Kiss, A. D. Ames, and G. Orosz, "Safety-critical control with input delay in dynamic environment," *IEEE Trans. Control Syst. Technol.*, vol. 31, no. 4, pp. 1507–1520, 2023.
- [34] S. Chen, Y. Sun, D. Li, Q. Wang, Q. Hao, and J. Sifakis, "Runtime safety assurance for learning-enabled control of autonomous driving vehicles," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2022, pp. 8978–8984.
- [35] S. Shalev-Shwartz, S. Shammah, and A. Shashua, "On a formal model of safe and scalable self-driving cars," arXiv preprint arXiv:1708.06374, 2017. [Online]. Available: <https://arxiv.org/abs/1708.06374>
- [36] A. Liniger and J. Lygeros, "Real-time control for autonomous racing based on viability theory," *IEEE Trans. Control Syst. Technol.*, vol. 27, no. 2, pp. 464–478, 2019.
- [37] E. Leurent, "An environment for autonomous driving decision-making," GitHub repository, 2018, version 1.12. [Online]. Available: <https://github.com/Farama-Foundation/HighwayEnv>
- [38] Q. Li, Z. Peng, L. Feng, Q. Zhang, Z. Xue, and B. Zhou, "MetaDrive: Composing diverse driving scenarios for generalizable reinforcement learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 3, pp. 3461–3475, 2023.
- [39] A. Yang *et al.*, "Qwen3 technical report," arXiv preprint arXiv:2505.09388, 2025. [Online]. Available: <https://arxiv.org/abs/2505.09388>
- [40] G. Kou *et al.*, "PADriver: Towards personalized autonomous driving," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, 2025, pp. 1–8.